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Basic Networking Commands

AIM:-

Study of various Network commands used in Linux and windows.

Procedure:-

1. Start the computer & login to the windows / Linux operating system.
2. Open the PUTTY application on the windows system.
3. Enter the IP address of the Linux Server and establish an SSH connection.
4. Log in using the valid username and password.
5. Execute the required N/W commands
6. Observe the O/P displayed for each command.

Description:-

commands I/P & O/P:-

① arp -a

O/P: Displays IP & MAC address table.

Observed: ARP cache entries were displa successfully.

② hostname

O/P: DESKTOP-ABC123

observed: The system hostname was displayed

(3) ipconfig

o/p: IP address, subnet Mask, Gateways, DNS.

observed:- Network configuration details were displayed.

(4) nbtstat -a <hostname>

o/p: NetBIOS name table

observed:- active network connections & routing information were displayed.

(5) net stat -r

o/p: Routing table

observed: routing & connections info was displayed.

(6) nslookup www.google.com

o/p: DNS server and IP address

observed: DNS name resolution was verified successfully.

(7) Pathping www.google.com

o/p: hop details & packet loss

statistics were analysed.

(8) ping www.google.com

O/P: Reply from IP with time and TTL

Observed: Network connectivity was verified successfully.

(9) route Print

O/P: Routing table entries

Observed:- The system routing table was displayed.

(10) ip address show

O/P:- Interface details with IP address

Observed: IP address assigned to the network interface was displayed.

Result:- 5/8/26

The Basic networking commands were executed successfully.